

```
-- single line comment
/*
multi
line
comments
*/
```

```
/*sql:-
(structured query language)
is a language that is used to communicate with databases.
Using sql you can
- create databases
- Create tables
- insert data
- update data
- delete data
- retrieve data
```

- Type of SQL Commands

1) DDL(Data definition language):-

DDL commands are used to define database structure.

- create
- alter
- drop
- truncate
- rename

example : create database databaseName;

2) DML(Data manipulation language):-

DML is used to modify data inside our tables

- insert
- update
- delete
- select

3) DQL(Data query language):-

used to fetch data

- Select

4) DCL(Data control language):-

used to control access/permissions

- grant
- revoke

5) TCL(Transaction control language):-

used to manage transactions

- commit
- rollback
- savepoint

```

*/

-- ddl commands
-- create :-
-- how to create database
create database testDB3;
create database if not exists testDB3;

-- how to use a database
use TESTDB3;

-- HOW TO CREATE TABLES
CREATE TABLE TABLE_NAME(COLUMN1 INT, COLUMN2 VARCHAR(225));
CREATE TABLE EMPLOYEES(E_Id INT, E_NAME varchar(225));

-- HOW TO SEE THE STRUCTURE OF YOUR TABLE
DESC TABLE_NAME;
DESCRIBE TABLE_NAME;

-- how to add a column to your table.
alter table employees add column salary decimal(9,2);
desc employees;
-- how to change a column name
alter table employees rename column salary to e_salary;

-- how to modify a columns data type
alter table employees modify e_salary varchar(225);

-- drop
-- how to drop a database
drop database testdb;
drop database testdb2;

-- how to drop a table
drop table if exists table_name;

--
desc employees;
-- how to drop a column from a table.
alter table employees drop column e_salary;

-- truncate:-
-- There are two to delete all the records from your table delete/ truncate
-- truncate is used to delete all the records from your table at once.
-- truncate is faster than delete.
-- truncate can't be rolled back.

```

```
insert into employees values(1,"Ray"),(2,"Shree"),(3,"Jay");
select * from employee_table;
desc employees;
```

```
truncate employees;
delete from employees;
```

```
-- Rename:-
-- how to rename a table
rename table employees to employee_table;
-- how to rename a column
alter table employee_table rename column e_id to emp_id;
desc employee_table;
```

```
/*
```

```
2) DML(Data manipulation language):-
```

```
DML is used to modify data inside our tables
```

```
- insert
- update
- delete
- select
```

```
*/
```

```
-- 1st Insert:-
```

```
-- 1st step
```

```
create database if not exists session1;
```

```
-- 2nd step
```

```
use session1;
```

```
-- 3rd step
```

```
create table employees (emp_id int, emp_name varchar(225), emp_age int);
```

```
-- 4th step
```

```
insert into employees values(101,"Raj",25);
```

```
insert into employees values
```

```
(102,"Sham",34),
```

```
(103,"Sonal",23),
```

```
(104,"Paras",45);
```

```
insert into employees values
```

```
(105,"Kiran",23),
```

```
(106,"Ravi",null);
```

```
insert into employees
```

```
(emp_name, emp_id) values
```

```
("Tisha",107);
```

```
insert into employees
```

```

(emp_age, emp_id, emp_name) values
(24,108,"Sahil"),
(34,104,"Kiran");
-- 5th step
select * from employees;
desc employees;
-----

-- 2) Update:-
select * from employees;

update employees
set emp_age = 30
where emp_id = 106;

-----

-- 3) delete:-
delete from employees where emp_id = 108;
select * from employees;
-----

-- Data types:-
-- 1) Numeric data types:-
-- 2) String data types:-
-- 3) Date and time data types:-
-- 4) Binary data types:-

-- 1) Numeric data types:-
-- Used to store numeric values:-
-- a) Integer types:-(stores whole numbers(no decimals))
-- Type          storage          range
-- tinyint        1 byte          -128 to 127    --> 0 - 255 unsigned
-- smallint        2 bytes          -32k to 32k
-- mediumint       3 bytes          -8m to 8m
-- int/Integer     4 bytes          -2B to 2B
-- bigint          8 bytes          very large number

create table testTable(testId tinyint);
create table testTable(testId tinyint unsigned);
insert into testTable values(255);
select * from testTable;

create table employees(emp_id int, emp_phone_number int);
alter table employees modify column emp_phone_number bigint;
insert into employees values(101,1234567890);
select * from employees;
insert into employees values(102,9816623423);
insert into employees values(102.967,9816623423);

```

```

-- Decimal/floating point numbers
-- used for numbers with decimals
-- 1) Decimal(Most important for money):-
desc employees;
alter table employees add column salary decimal(9,2);

select * from employees;
update employees set salary = 95000000.32 where emp_id = 103; -- 2.132 --> 2 --> 3

-- 2) Float and double

alter table employees modify column attendance double;

select * from employees;

update employees set attendance = 78.32423523432423324324
where emp_id = 103;

show variables like "%safe%";

set sql_safe_updates = off;
set sql_safe_updates = 0;
set sql_safe_updates = on;
set sql_safe_updates = 1;

-- -----
-- string data types
-- Char:-
-- is stores fixed length strings and occupy all the data that has been allocated
to it:-
create table students(s_id int, sname char(10));

insert into students values(1, "Karan");

-- varchar(length):-
-- varchar is used to store strings of variable length
alter table students modify column sname varchar(225);
insert into students values(102, "Shree");
desc students;

-- text:-
-- used for large text
-- types of text :- tinytext 255 char, text 65k char, mediumtext 16mb, longtext 4gb

-- enum :-
-- enum stores one value from a predefined list.
-- enum ensures valid values are inserted

```

```
create table employees2 (emp_name varchar(225), emp_degree
enum("BTech","BBA","MTech","MBA"));
insert into employees2 values("kashish","btech,mba");
select * from employees2;
```

```
-- set :-
-- set stores multiple values from a predefined list
create table employees2(emp_name varchar(225), emp_skills
set("Java","python","SQL","AWS","JS"));
insert into employees2 values("Ron", "java,sql,aws,Ruby");

select * from employees2;
```

```
-- date and time data types:-
```

```
-- -----
-- create a database
-- name the database session1
-- use the database
-- create a table "Employees" with columns emp_id int, emp_name varchar(225),
emp_salary decimal(9,2),
-- emp_attendance float, emp_cgpa float.
-- modify column emp_cgpa data type to double
-- change column emp_attendance to emp_attd
-- rename table employees to employees_demo
-- insert 5 records to the table
-- update 1 record and set their salary to 90000
-- delete 1st employee from your table
```